

SkillSphere Nexus

Enterprise Learning, Certification & Career Development Management Platform

8 Weeks | 4 Milestones

1. Project Idea

SkillSphere Nexus is a cloud-native enterprise platform designed to manage employee learning, certifications, training programs, career progression, competency frameworks, and workforce development. Instead of AI-generated career recommendations, the platform focuses on Learning Management, Certification Tracking, Employee Skill Management, Career Development Planning, Training Administration, Competency Frameworks, Internal Job Opportunities, Learning Analytics, Workforce Development, and Enterprise Talent Management. The platform uses Angular 20 frontend with Java 25 and Spring Boot 4 backend. PostgreSQL database with Redis cache. Microservices include User Service, Skill Service, Learning Service, Career Service, Training Service, and Reporting Service. Kafka enables event-driven architecture. Keycloak provides IAM. The system manages 12.4K employees, 2,847 skills, 847 courses with 87% completion, 8.4K active certifications with 94% renewal, and 2,847 career plans with 247 promotions annually.

2. Emerging Skills

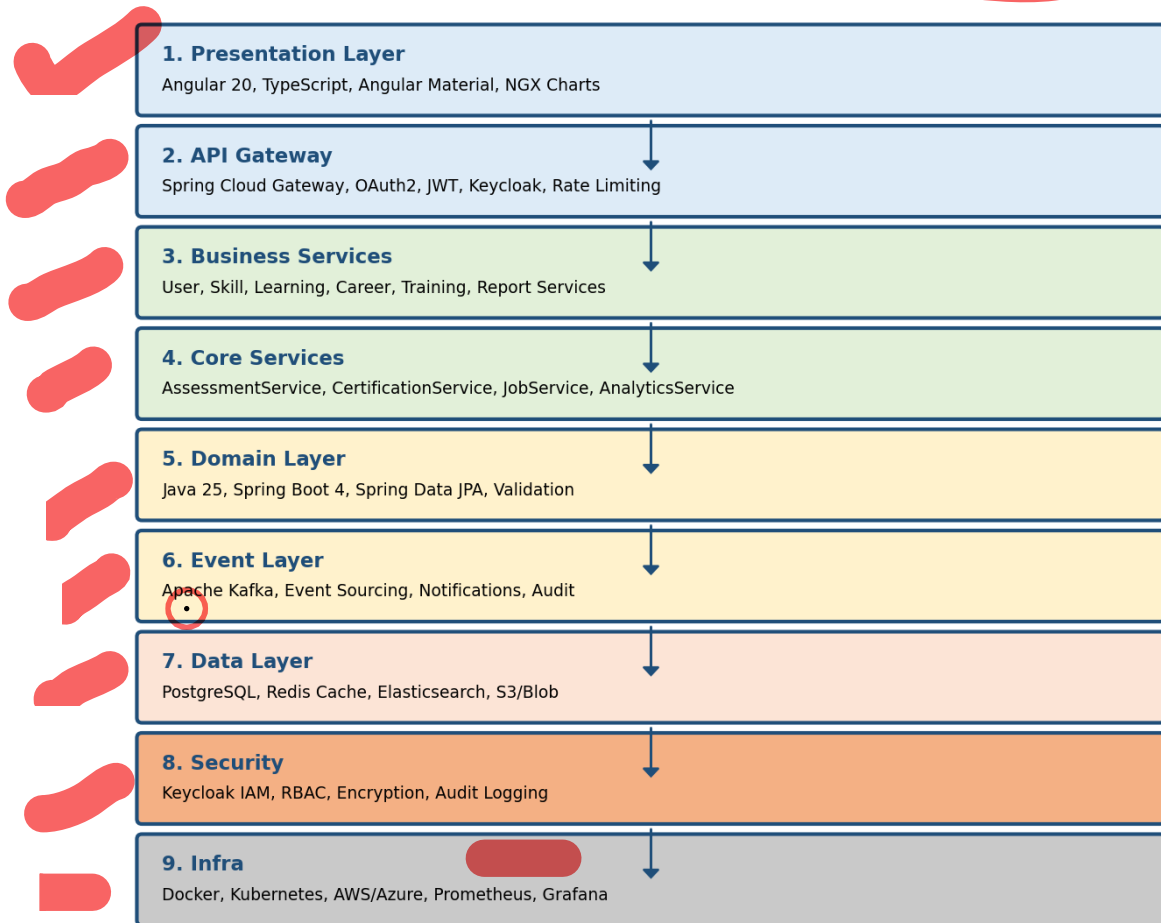
- Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

3. Technology Stack

Component	Technology
Frontend	Angular 20
Backend	Java 25, Spring Boot 4
Database	PostgreSQL
Cloud	AWS EC2, RDS, S3, or Azure App Service, PostgreSQL, Blob
Skills	Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

4. System Architecture Diagrams

SkillSphere Nexus - 9-Layer Enterprise Learning Architecture



9-layer architecture with Angular 20 frontend, Spring Cloud Gateway, Java 25/Spring Boot 4 microservices, Kafka event streaming, and PostgreSQL data layer.

5. Project Working Diagram

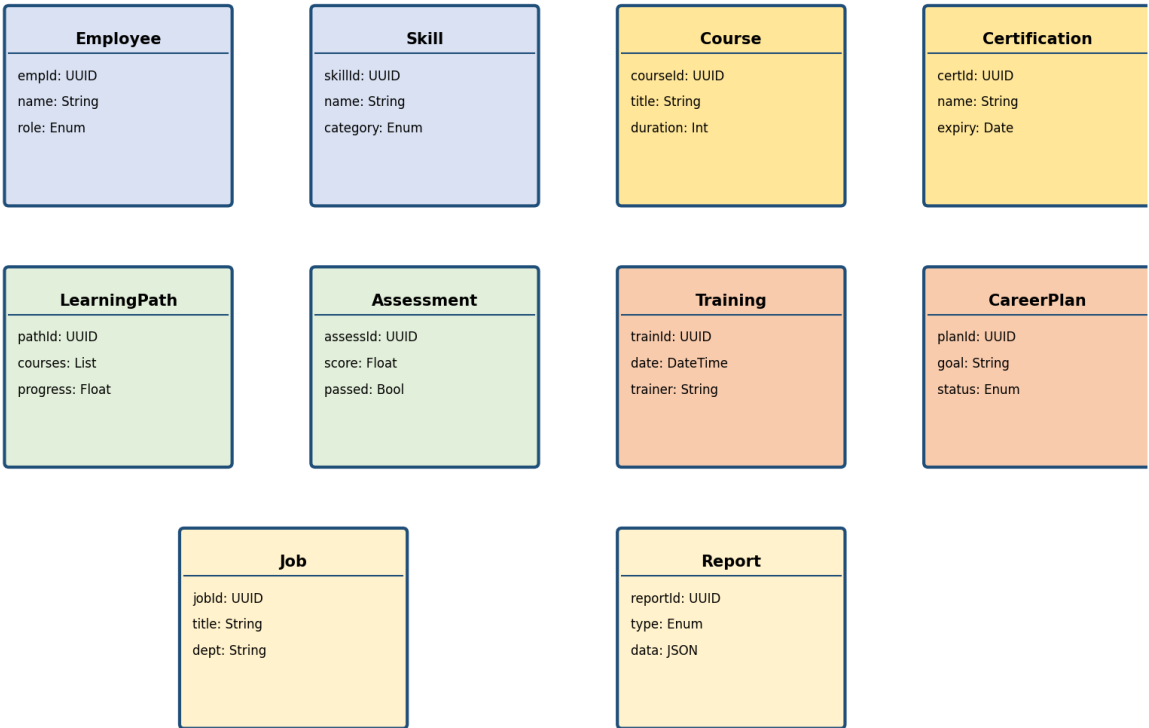
SkillsSphere Nexus - Learning & Career Development Workflow



Learning & career workflow: Employee Enrollment → Skill Assessment → Learning Paths → Course Completion → Certification → Career Planning → Analytics Dashboard.

6. Class Diagrams

SkillsSphere Nexus - Class Diagram: Learning & Career Domain



Core entities: Employee, Skill, Course, Certification, LearningPath, Assessment, Training, CareerPlan, Job, Report.

7. Milestone-wise Deliverables & Screens

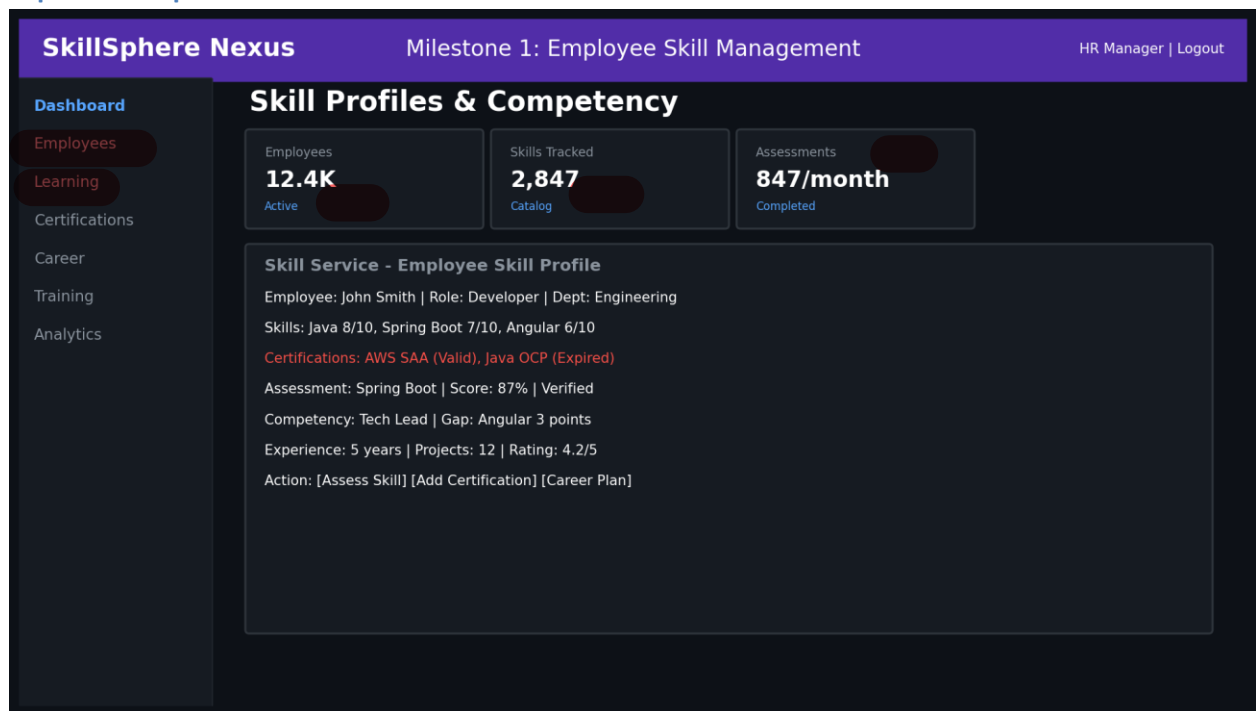
Milestone 1 (Weeks 1-2) – Employee Skill Management

Implements employee skill profiles and competency tracking. Manages 12.4K employees. Tracks 2,847 skills across technical, domain, and soft skills. Provides skill assessments with verification. Maps competency frameworks. Tracks certifications and experience levels.

Key Modules

- Skill Service
- Employee Profiles
- Skill Catalog
- Assessments
- Competency Framework
- Certification Tracking

Expected Output Screen



Skill Service: John Smith, Developer. Skills: Java 8/10, Spring Boot 7/10. AWS SAA valid, Java OCP expired. Assessment: 87%.

Validation Screens

- Employee profile
- Skill catalog

- Assessment scoring
- Competency mapping
- Certification tracking
- RBAC by HR

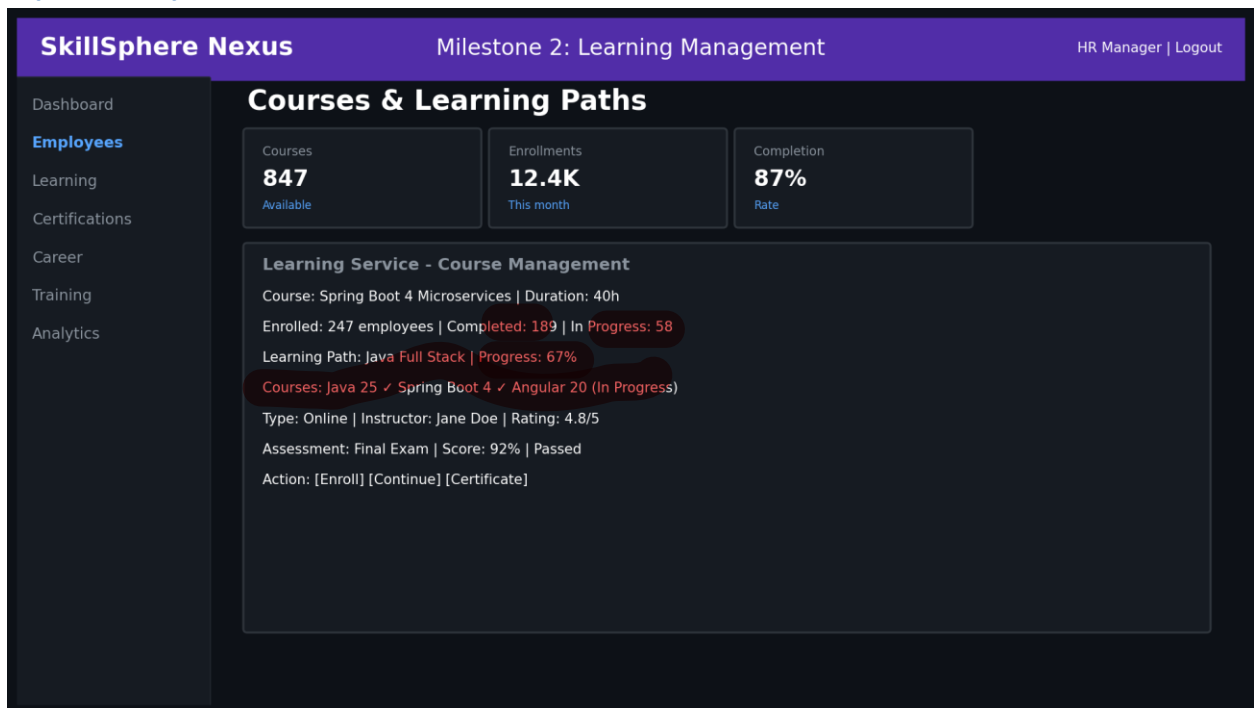
Milestone 2 (Weeks 3-4) – Learning Management

Deploys a learning management system. Offers 847 courses. Handles 12.4K enrollments monthly. Achieves 87% completion rate. Provides learning paths for career tracks. Supports online courses, workshops, webinars, and bootcamps. Tracks course completion.

Key Modules

- Learning Service
- Course Management
- Enrollment Tracking
- Learning Paths
- Course Completion
- Content Management

Expected Output Screen



Learning Service: Spring Boot 4 Microservices. 247 enrolled, 189 completed. Learning Path: Java Full Stack 67%. Score: 92%.

Validation Screens

- Course creation

- Enrollment tracking
- Learning path
- Completion tracking
- Assessment results
- Certificate generation

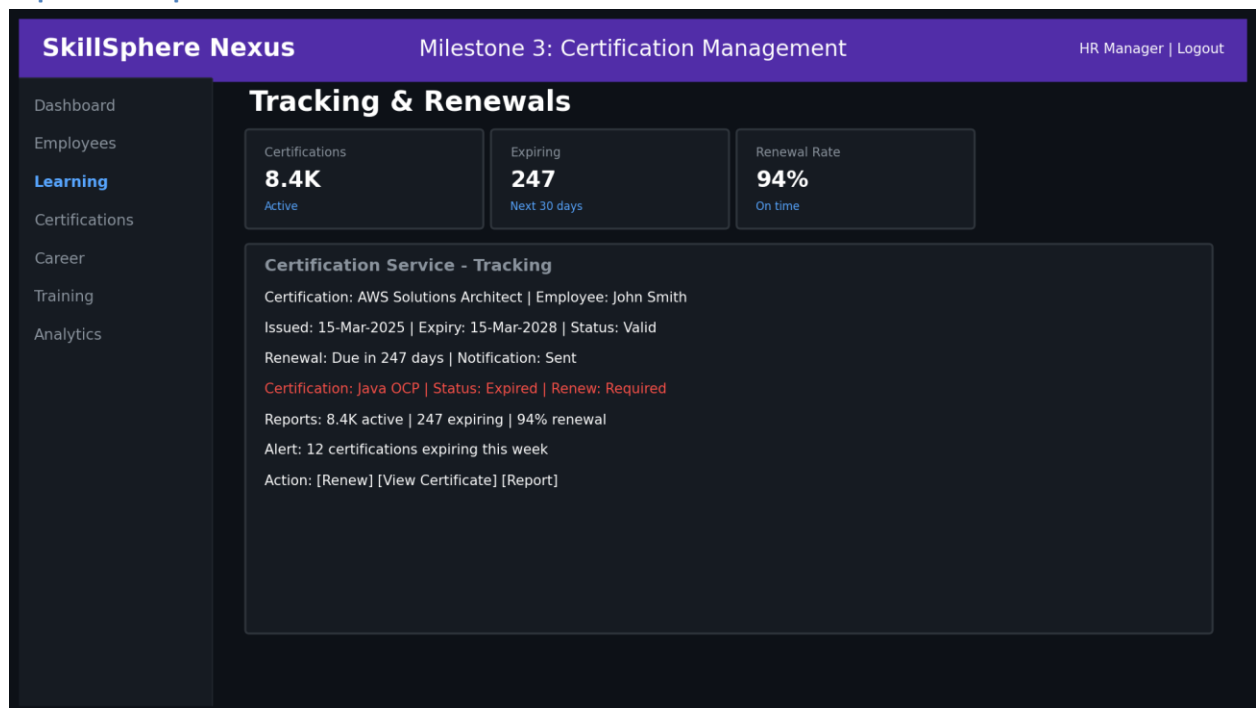
Milestone 3 (Weeks 5-6) – Certification Management

Implements certification registration and tracking. Manages 8.4K active certifications. Sends renewal notifications. Achieves 94% renewal rate. Tracks expiry dates. Generates certification reports. Monitors 247 expiring in the next 30 days.

Key Modules

- Certification Service
- Expiry Tracking
- Renewal Notifications
- Certification Reports
- Compliance Tracking
- Audit Trail

Expected Output Screen



Certification Service: AWS SAA valid until 15-Mar-2028. Java OCP expired. 8.4K active, 247 expiring. 94% renewal rate.

Validation Screens

- Certification registration
- Expiry tracking
- Renewal notifications
- Compliance verification
- Report generation
- Audit trail

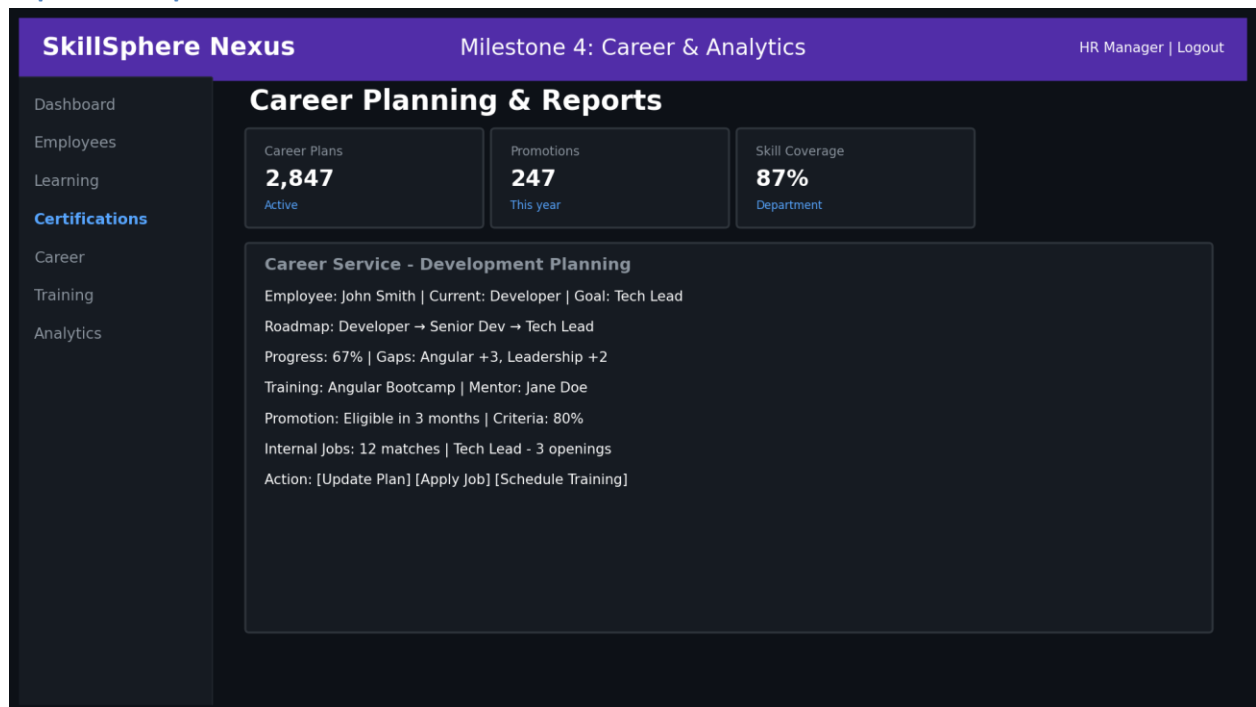
Milestone 4 (Weeks 7-8) – Career & Analytics

Deploys career development planning and analytics. Manages 2,847 active career plans. Facilitates 247 promotions annually. Tracks skill coverage at 87% department level. Provides training effectiveness reports. Offers an internal job portal with 12 matches.

Key Modules

- Career Service
- Career Roadmaps
- Promotion Criteria
- Internal Job Portal
- Training Analytics
- Executive Dashboard

Expected Output Screen

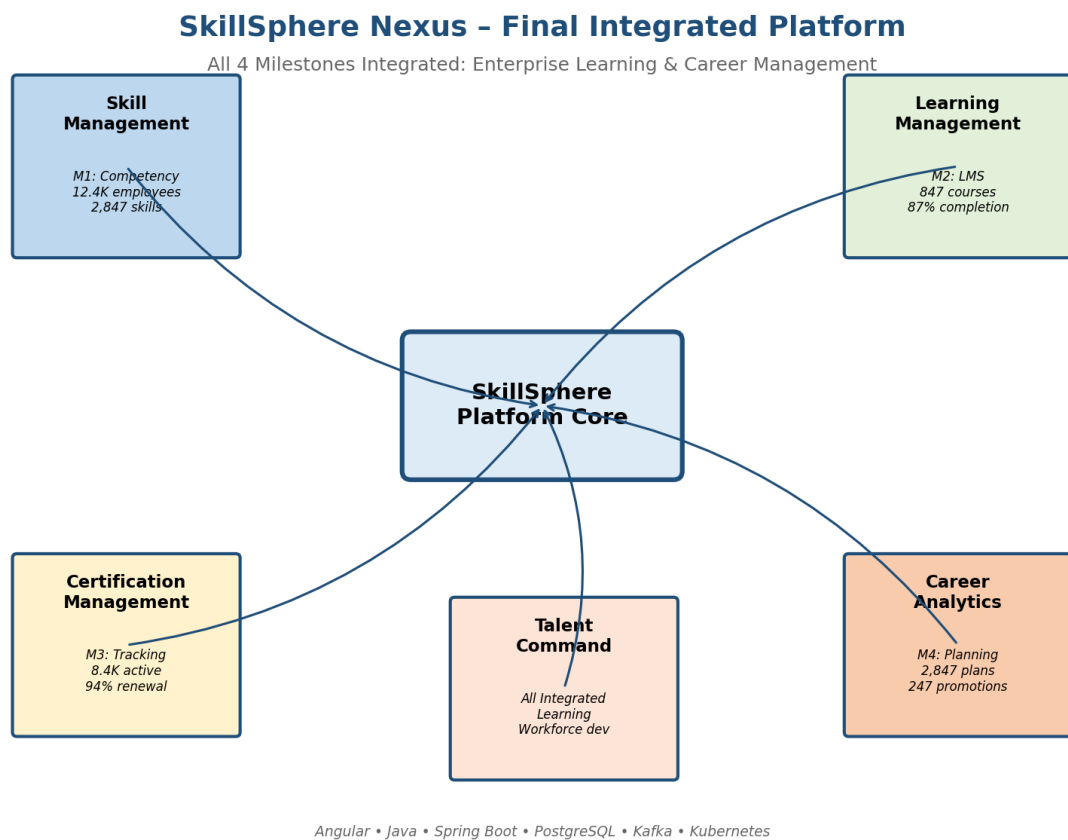


Career Service: John Smith, Developer to Tech Lead. Progress 67%. Gaps: Angular +3. Mentor: Jane Doe. 12 job matches.

Validation Screens

- Career roadmap
- Promotion criteria
- Skill gap analysis
- Training recommendations
- Job matching
- Progress tracking

8. Master Application Screens – Final Integrated



Final integrated master screens after all 4 milestones:

- Skill Management: M1 – Competency, 12.4K employees, 2,847 skills, assessments
- Learning Management: M2 – LMS, 847 courses, 87% completion, learning paths
- Certification Management: M3 – Tracking, 8.4K active, 94% renewal, 247 expiring
- Career Analytics: M4 – Planning, 2,847 plans, 247 promotions, 87% coverage
- Talent Command: All integrated – Learning, workforce development, enterprise talent

9. Validation Screens Framework

All screens enforce: Employee skill validation, Course enrollment tracking, Certification expiry monitoring, RBAC by admin/HR manager/training manager via Keycloak, and comprehensive audit logging for all learning actions. Operations show skill levels + course progress + certification status. Critical actions require approval. Kafka ensures event ordering. All activities are logged immutably. Skills validated. Learning tracked. Careers planned.